CYLINDER HEAD COVER (Continued)

(4) Install cylinder head cover assembly to cylinder head. Install all bolts, ensuring the two (2) bolts containing the sealing washer are located in the center locations of cover. Tighten bolts in sequence shown in (Fig. 27). Using a 3 step torque method as follows:

(a) Tighten all bolts to 4.5 N·m (40 in. lbs.).
(b) Tighten all bolts to 9.0 N·m (80 in. lbs.).
(c) Tighten all bolts to 12 N·m (105 in. lbs.).

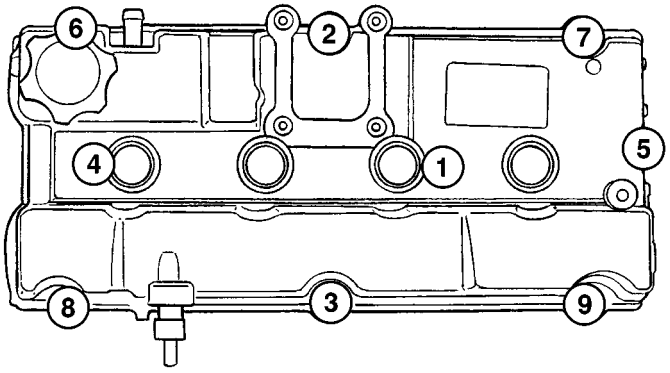

Fig. 27 CYLINDER HEAD TIGHTENING SEQUENCE

(5) Install intake manifold.

(6) Install ignition coil and spark plug wires. Tighten fasteners to 12 N·m (105 in. lbs.).

(7) If the PCV valve was removed, apply Mopar® Thread Sealant with Teflon to threads and install valve to cylinder head cover. Tighten PCV valve to 8 N·m (70 in. lbs.).

(8) Connect PCV and make-up air hoses to cylinder head cover.

INTAKE/EXHAUST VALVES & SEATS

DESCRIPTION

The four valves per cylinder are opened by using roller rocker arms which pivot on hydraulic lash adjusters. The valves have chrome plated valve stems. Viton rubber valve stem seals are integral with the spring seats. They have chrome plated stems to prevent scuffing. Viton rubber valve stem seals are integral with the spring seats. The valves, spring retainers,and locks, are the 3 - bead lock design

CLEANING

(1) Clean all valves thoroughly and discard burned, warped and cracked valves.

VALVE SPRINGS

REMOVAL

REMOVAL - CYLINDER HEAD ON

(1) Remove camshafts.

(2) Rotate crankshaft until piston is at TDC on compression.

(3) With air hose attached to adapter tool installed in spark plug hole, apply 90-120 psi air pressure.

(4) Using Special Tool MD-998772-A with adapter 6779 (Fig. 28), compress valve springs and remove valve locks.

(5) Remove valve spring(s).

(6) Remove valve stem seal(s) by a using valve stem seal tool (Fig. 30).

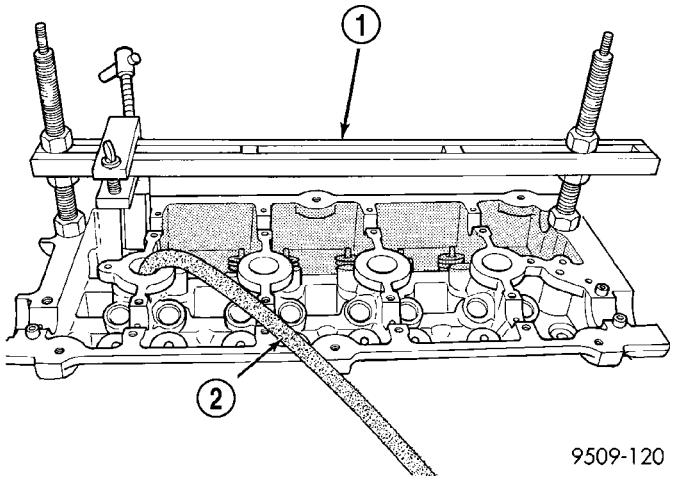

Fig. 28 Valve Spring - Removal/Installation

1 - VALVE SPRING COMPRESSOR MD 998772A
2 - AIR HOSE

REMOVAL - CYLINDER HEAD OFF

(1) With cylinder head removed from cylinder block, compress valve springs using a universal valve spring compressor.

(2) Remove valve retaining locks, valve spring retainers, valve stem seals and valve springs.

(3) Before removing valves, **remove any burrs from valve stem lock grooves to prevent damage to the valve guides.** Identify valves, locks and retainers to insure installation in original location.

(4) Inspect the valves. (Refer to 9 - ENGINE/CYLINDER HEAD/VALVE SPRINGS - INSPECTION)